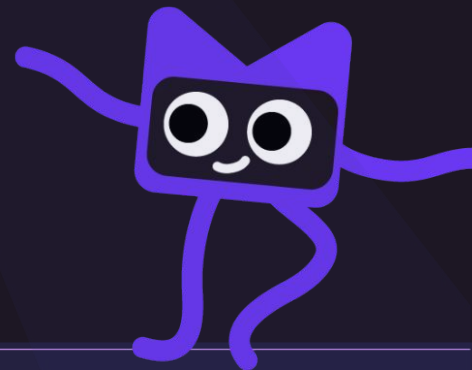




On in-database analytics

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About Me

- Clojure community organizer
- IT Consultant, speaker, writer
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| 從錯誤到創新

從錯誤到創新

跨領域的錯誤處理、創新之道

https://leanpub.com/errors_to_innovation/

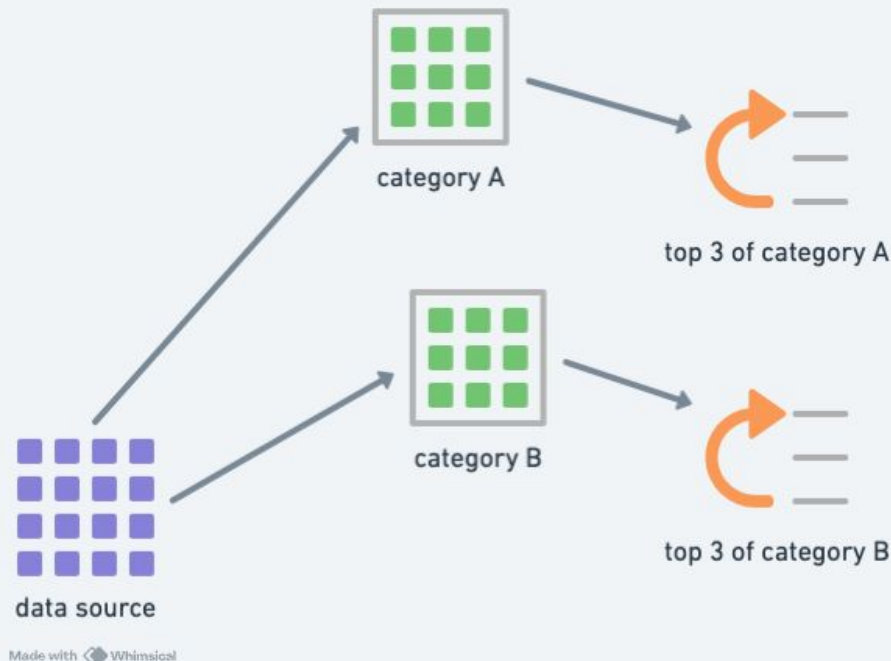


Agenda

- The story about **Top n Per Group**
- The value of in-database analytics
- The tooling of in-database analytics
- If you don't like it, ...
- Conclusion

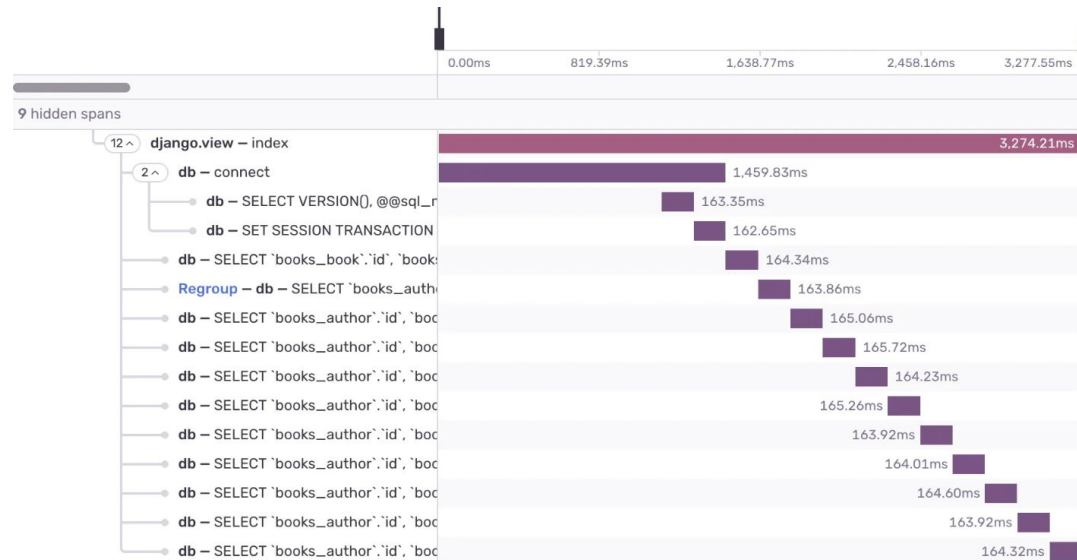
Top n Per Group - definition

- Years ago, I encountered it.
 - Top n Per Group



Top n Per Group - N+1 query problem

- A Colleague wrote N+1 queries.
=> **N+1 query problems.**
 - Inconsistent if writing happens
 - I/O delay



Top n Per Group

country	city	population
United States	New York	8175133
United States	Los Angeles	3792621
United States	Chicago	2695598
France	Paris	2181000
France	Marseille	808000
France	Lyon	422000
United Kingdom	London	7825300
United Kingdom	Birmingham	1016800
United Kingdom	Leeds	770800

Top n Per Group - window function

- Years later, I discovered [window function](#).
- What is **window function**?



```
SELECT
  *
FROM
  (
    SELECT
      country,
      city,
      population,
      row_number() OVER (
        PARTITION BY country
        ORDER BY
          population desc
      ) AS country_rank
    FROM
      cities
  ) ranks
WHERE
  country_rank <= 2;
```

The value of in-database analytics

- Query composition to ensure consistency
- Reduce I/O to boost performance
- Declarative semantic

In-database analytics: Reduce I/O

- Definition: In-database analytics is a technology that allows data processing to be conducted within the database by **building analytic logic into the database** itself. Doing so eliminates the time and effort required to transform data and move it back and forth between a database and a separate analytics application.

The toolings of in-database analytics

- Subquery and window function
- Data Engineering
 - dbt/Jinja
- Data Science
 - neo4j
 - Spark
 - Apache MADlib

(Data Engineering) dbt/Jinja

- problem:
SQL is not composable.
- solution:
We use Jinja template lang
to make it composable.

```
select
    attribute_1,
    attribute_2,
    attribute_3,
    attribute_4,
    attribute_5
from table_1
```

(Data Engineering) dbt/Jinja

```
select
```

```
    {% for i in range(1,6) -%}  
        attribute_{{ i }} {%- if not loop.last %},{% endif %}  
    {% endfor %}
```

```
from table_1
```

(Data Engineering) dbt/Jinja - data lineage graph

Lineage Graph



resources All selected packages jaffle_shop tags untagged --select ... --exclude ...

Update Graph



(Data Science) Spark/neo4j/Apache MADlib

- Shortest Path (路徑)
- Page Rank
- Inverse Matrix
- k-means clustering (分群)
- Linear Regression

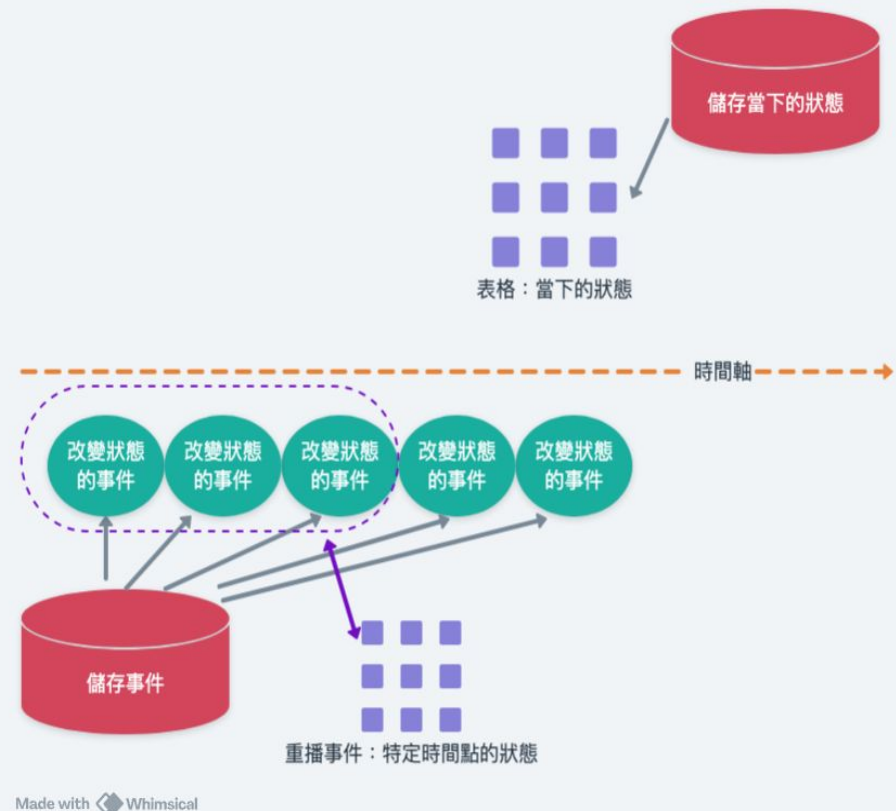
If you don't like in-database analytics ...

- Don't do data engineering jobs.
- Use Datomic.

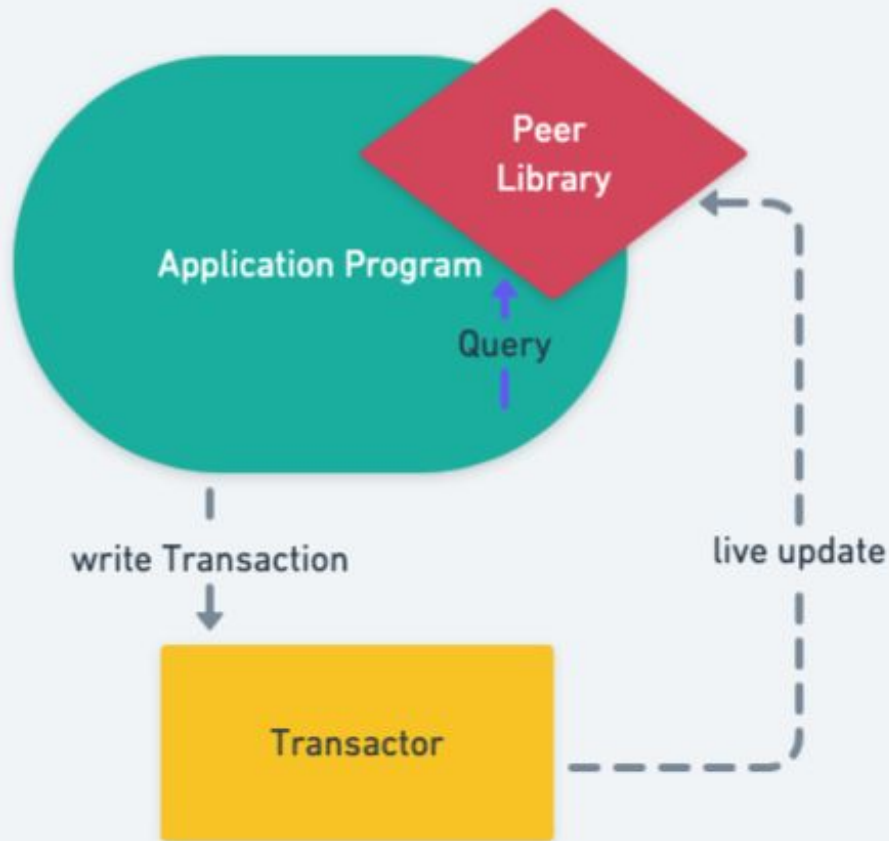


Event sourcing in Datomic

- Database as value
- Always Consistent
- Every Datom consists of `E / A/ V/ T/ op`
 - E - entity
 - A - attribute
 - V - value
 - T - transaction ID
 - op - assert, retract



Querying in application



Made with Whimsical

- In Datomic, query is running inside the Peer Library.
- Not so much I/O saved.

Top N per Group using Datomic's query (typical way)

- Given that there is no big I/O penalty and no inconsistent issues, we just use query to handle **group-by**, and use application program to handle **the rest**.



Top N per Group using Datomic's query

UDA

- user-defined aggregate

Declarative

Clean and Succinct.

- compared to SQL
- compared to non-UDA

```
(defn max-by-population
  [coll] ← coll - [{name, population}, ...]
  (let [result (sort-by second > coll)]
    (take 2 result)))

(d/q '[:find ?country (repl-sessions.day18/max-by-population ?tup)
      :in $
      :where
      [?ct :country/name ?country]
      [?ct :city/name ?city]
      [?ct :city/population ?population]
      [(tuple ?city ?population) ?tup]]
  (db/db))
```

Conclusion: the value of in-database analytics

- Avoid **inconsistent state**. (OLTP use case with RDBMS)
- Reduce I/O to **boost performance**.
- Provide **clean and succinct semantic** with the help of UDA (user-define aggregate).

Reference

- 當代資料工程與資料分析
<https://ithelp.ithome.com.tw/users/20161869/ironman/6057>
- Datomic, 內建事件溯源的資料庫
<https://ithelp.ithome.com.tw/users/20161869/ironman/7432>